

Robust Window and Exact-Pressure Optimization for Simple Zeros of the Riemann Zeta Function

AMTOPA

August 12, 2026

Abstract

We record a computer-assisted research-draft refinement of the positioned-pressure method for simple zeros of the Riemann zeta function on the critical line. The refinement retains exact shifted-block pressure accounting, uses a robust 15-term analytic window, and closes the resulting six-gap local inequality by outward-rounded interval branch-and-bound. Subject to the imported analytic interface and finite-dimensional Gram profile used by the predecessor lineage, the current certified numerical inputs yield

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.67333146637444245098 \dots,$$

i.e. 67.3331466374% before conservative decimal truncation. This manuscript is a research draft, not a peer-reviewed theorem and not a proof of the Riemann hypothesis.

1 Imported analytic interface

Let $S = N_0^s(T, 2T)$ and $N = N(T, 2T)$. We import

$$S \geq HN + \Delta(M) - o(N)$$

and the finite- m Gram profile

$$h_m(E) = \begin{cases} E, & 0 \leq E \leq m/(m-1), \\ E/m + 2\sqrt{(m-1)E/m} - 1, & E \geq m/(m-1). \end{cases}$$

These ingredients belong to the Anthropic / `trmdy` / `sxuff` lineage and are not reproved here.

2 Robust 15-term window

We use

$$v(s) = \sum_{j=0}^{14} c_j \cos(\omega_j s), \quad \omega_0 = \sqrt{2}, \quad \omega_j = 2j\pi \ (1 \leq j \leq 14),$$

with exact rational coefficients of common denominator 10^9 and numerator vector

$$(1000000000, 8629738, -10085378, 1746803, 1125700, -2203905, 1904615, \\ -4559603, 7930665, -3022627, 2165339, 398121, -255934, 188899, -148305).$$

Direct interval arithmetic encloses

$$H(v) = 0.67233388665794215883536822355662 \dots$$

and verifies the working rational floor

$$H_{\text{cert}} = \frac{3361669433}{5000000000} = 0.6723338866.$$

The same interval calculation also verifies positivity of the window on the required support.

3 Local seven-point certificate

The exact pair weights are recorded in `candidate.json`. The position-pressure coefficients are

$$\frac{1}{2300000000}(831522, 1096590, 1071888, 1071888, 1096590, 831522),$$

so their exact sum is

$$B = \frac{3}{1150}.$$

The robust floating-point search for the retained rational window observed a lowest basin near

$$0.005402429240910082\dots$$

A shared-table interval ladder certified 0.0054015, 0.0054020, and 0.0054022. The next tested values 0.0054024 and 0.00540242 were inconclusive at a terminal cell and are not claimed. Thus the promoted rigorous local value is

$$\varepsilon = \frac{27011}{5000000} = 0.0054022.$$

The 4000-grid, 50-digit outward-rounded interval branch-and-bound run reports

$$\text{nodes} = 3380026, \quad \text{pruned} = 1690085, \quad \text{splits} = 1689941,$$

with 1980120 convex boxes, 852267 tangent prunes, and maximum depth 70. Exact table hashes and workflow provenance are recorded in `candidate.json` and `certificates/latest-verification.txt`. Independent reproduction remains requested.

4 Exact shifted-pressure averaging

For an m -point block, summing the seven-point inequality over the $m - 6$ local windows gives total pressure charge

$$(m - 6)B.$$

Across the shifted block partitions, exact double counting retains this factor rather than replacing it by a coarser $B(m - 1)$ bound.

Set

$$A_m = \varepsilon(m - 6), \quad R_m = h_m(A_m), \quad \eta_m = \frac{R_m}{A_m}.$$

Then the block-defect estimate gives

$$\Delta(M) \geq \frac{R_m}{m}S - \frac{\eta_m B(m - 6)}{m}N - o(N),$$

and hence

$$\boxed{\frac{S}{N} \geq \frac{mH_{\text{cert}} - \eta_m B(m - 6)}{m - R_m} - o(1).}$$

5 Final arithmetic

With

$$H_{\text{cert}} = 0.6723338866, \quad \varepsilon = 0.0054022, \quad B = 3/1150,$$

the integer scan selects $m = 204$. Here

$$A_{204} = 1.0696356,$$

$$R_{204} = 1.06863109716549584957366956134\dots,$$

and

$$\eta_{204} = 0.999060892481042936093067172916\dots$$

Therefore

$$\frac{S}{N} \geq 0.673331466374442450980484784486\dots - o(1),$$

and safely

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} > 0.6733314663.$$

6 Trust boundary and status

The exact pressure-position identities, interval window evaluation, local interval certificate, and final arithmetic are directly checkable from the repository. The explicit-formula / trace interface and the finite- m Gram profile remain imported from prior work. The local certificate has not yet been independently reproduced with a second implementation, so the percentage should be read as an interval-certified research-draft candidate rather than a publication-ready theorem.

Reproducibility

The repository contains exact parameters in `candidate.json`, arithmetic checks and the native verifier in `src/`, archived predecessor records under `archive/`, and a GitHub Actions workflow that compiles this manuscript whenever the L^AT_EX source changes.